

Ayan Baruah

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EDUCATION

Indian Institute of Technology (ISM) Dhanbad
Bachelor of Technology in Civil Engineering (GPA: 8.10)

Expected May 2028
Dhanbad, India

South Point School, Guwahati
Class XII (CBSE) - (Percentage: 90.2%)

2024
Guwahati, India

Don Bosco School, Guwahati
Class X (CBSE) - (Percentage: 97.2%)

2022
Guwahati, India

PROJECTS

CodeArena — Real-Time Competitive Coding Platform with Custom Contest Rooms | [MERN](#), [JavaScript](#) | [GitHub](#)

- Developed a full-stack coding contest platform using the MERN stack, featuring real-time **Socket.IO-based** contest rooms where users can create custom contests, compete with friends, and interact via live chat.
- Engineered a secure and scalable code execution pipeline using **Docker-based sandboxing**, ensuring isolated and safe execution of user submissions.
- Implemented asynchronous job processing with **BullMQ and Redis** to efficiently handle high-volume concurrent submissions.
- Implemented real-time **collaborative code editing** using **Yjs CRDTs** and Socket.IO, enabling **Google Docs-style live cursor tracking**, awareness indicators, and conflict-free multi-user synchronization within a Monaco-based code editor.

Trivalent Job Board — MERN, TypeScript, OpenAI API | [GitHub](#) **Honourable Mention — AI Impact Summit Hackathon 2026**

- Developed a full-stack recruitment platform connecting job seekers and employers with **AI-driven job matching** and **automated resume parsing**.
- Integrated **OpenAI API** to extract skills from PDF resumes and recommend relevant job roles.

Road SOS — Next.js, React, TypeScript | [Live Demo](#) | [GitHub](#)

- Developed and deployed a cross-platform emergency web-app, significantly reducing accident response time by **automating SOS triggers** and routing users to the nearest hospitals and police stations.
- Engineered a Capacitor Motion G-force collision detector instantly **transmitting the victim's live location to emergency contacts**, with a 10s cancel buffer and offline native SMS fallback.
- Optimized geospatial rendering by caching nearby resources in localStorage (skipping API fetches unless moved 500m via the Haversine formula), integrating the OSRM API to overlay real-time optimal driving routes.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, HTML5

Frameworks: React.js, Next.js, Express.js, Node.js, Tailwind CSS

Databases: MongoDB, MySQL, PostgreSQL, Redis

Tools: Git/GitHub, Postman, Firebase, Docker, Kubernetes, BullMQ, VS Code

Course Works: Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems

Interests: Full Stack Development, Competitive Programming, Machine Learning, Generative AI, Quant Finance

ACHIEVEMENTS

- Codeforces Rating: 1408 (Specialist) [Codeforces Profile](#)
- Solved 300+ Data Structures and Algorithms problems on LeetCode, Codeforces.
- Strong foundation in data structures, algorithms, and performance-optimized problem solving.

EXTRA-CURRICULAR ACTIVITIES

Sports: Badminton (State-level) | Represented IIT (ISM) Dhanbad at **Inter IIT Sports Meet 2024 & 2025**